this convention whenever a result requires a single coordinate; the exchange principle and Lemma 2.15 below are false for arbitrary tuples.

From here on we work in a sufficiently saturated model $\mathbb{U}$ of $\mathrm{ACVF}(0,0)$, together with a parameter set $\mathbb{S}$, which is supposed to be a substructure of $\mathbb{U}$. So by a definable set we shall mean an $\mathbb{S}$-definable set, unless indicated otherwise. For the rest of this paper, we assume that $\Gamma(\mathbb{S})$ is nontrivial.

Notation 2.2. There is a special element $\infty = \mathrm{rv}(0)$ in the RV-sort. For simplicity, we shall write RV to mean the RV-sort without the element ∞, and RV_∞ otherwise — that is, $\mathrm{RV} = \mathrm{rv}(\mathrm{VF}^\times)$ and $\mathrm{RV}_\infty = \mathrm{rv}(\mathrm{VF})$ — although the difference rarely matters (when it does we will of course provide further clarification).

Terminology 2.3 (Sets and subsets). By a definable set in VF we mean a definable subset in the sort VF, by which we just mean a subset of VF^n for some n, unless indicated otherwise; similarly for other sorts or structures in place of VF that have been clearly understood in the context, such as RV_∞, $\Bbbk$, $\mathcal{M}$, or any substructure $\mathbb{M}$ of $\mathbb{U}$. In particular, a definable set without further qualification means a definable set in $\mathbb{U}$, that is, a definable subset of $\mathrm{VF}^n \times \mathrm{RV}_\infty^m$ for some $n, m \in \mathbb{N}$.

Terminology 2.4 (Relations as functions). We may think of a relation C between two sets A and B, that is, a subset $C \subseteq A \times B$, as a function from A into the powerset $\mathcal{P}(B)$. For instance, if $A \subseteq \mathrm{VF}^n \times \mathrm{RV}^m$ is a definable set then it may be thought of as a *definable* function from A_{VF} into $\mathcal{P}(\mathrm{RV}^m)$. Thus a definable function $f : A \longrightarrow \mathcal{P}(B)$ is ontologically the same as the definable set $\bigcup_{a\in A} a \times f(a)$. We say that f is *finitary* if $f(a)$ is finite for every $a \in A$.

The default topology in $\mathbb{U}$ is the valuation topology.

Notation 2.5. A set $\mathfrak{p} \subseteq \mathrm{VF}^n \times \mathrm{RV}_\infty^m$ of the form $\prod_{i\le n} \mathfrak{b}_i \times t$ is called an (*open, closed,* RV-) *polydisc* if each $\mathfrak{b}_i$ is an (open, closed, RV-) disc. The *radius* $\mathrm{rad}(\mathfrak{p})$ of the polydisc $\mathfrak{p}$ is the tuple $(\mathrm{rad}(\mathfrak{b}_1), \ldots, \mathrm{rad}(\mathfrak{b}_n))$. The open polydisc centered at $(a,t) \in \mathrm{VF}^n \times \mathrm{RV}_\infty^m$ with radius $\gamma \in \Gamma^n$ is denoted by $\mathfrak{o}((a,t),\gamma)$; the closed polydisc with radius $\gamma \in \Gamma_\infty^n$ is denoted by $\mathfrak{c}((a,t),\gamma)$. Radius ∞ is used only for closed discs, with $\mathfrak{c}(a,\infty) = \{a\}$; open discs have finite valuative radius.

The subdiscs $\mathfrak{o}(0,\gamma)$, $\mathfrak{c}(0,\gamma)$ of VF are more suggestively denoted by $\mathcal{M}_\gamma$, $\mathcal{O}_\gamma$, respectively, and $\mathcal{M}_0$, $\mathcal{O}_0$ are even more suggestively denoted by $\mathcal{M}$, $\mathcal{O}$. Write $\mathrm{RV}_\infty^{\circ\circ} = \mathrm{rv}(\mathcal{M})$ and $\mathrm{RV}^{\circ\circ} = \mathrm{RV}_\infty^{\circ\circ} \smallsetminus \infty$. More generally, for $\gamma \in \Gamma$, denote by $\mathrm{RV}_\gamma^{\circ\circ}$ the set $\mathrm{rv}(\mathcal{M}_\gamma \smallsetminus 0)$.

An RV-*disc* is a fiber $\mathrm{rv}^{-1}(t)$ for $t \in \mathrm{RV}_\infty$, including $\mathrm{rv}^{-1}(\infty) = \{0\}$. In a mixed ambient space, an RV-polydisc keeps the existing RV_∞-coordinates fixed. The RV-*hull* of a set A, denoted by $\mathrm{RVH}(A)$, is the union of all the RV-polydiscs whose intersections with A are nonempty. If A equals $\mathrm{RVH}(A)$ then A is called an RV-*pullback*.

Resolved referee R022 [Clarification] – RV-discs and the zero fiber.
Original source line 748. Define an RV-disc explicitly as $\mathrm{rv}^{-1}(t)$ for $t \in \mathrm{RV}_\infty$, including $\mathrm{rv}^{-1}(\infty) = \{0\}$. This convention is needed for the equivalence between being an RV-pullback and having empty RV-boundary; otherwise sets meeting coordinate hyperplanes are not covered by their stated RV-hulls. Also specify that taking the hull keeps the existing RV-coordinates fixed.

Author resolution. This is the intended convention. The definition now explicitly includes the singleton zero fiber and preserves the existing RV-coordinates under the hull.

Terminology 2.6. A VF-*fiber* of a set A is a set of the form A_t, where $t \in A_{\mathrm{RV}}$; in particular, a VF-fiber of a function $f : A \longrightarrow B$ is a set of the form f_t for some $t \in f_{\mathrm{RV}}$ (here f also stands for its own graph) which is indeed (the graph of) a function. We say that A is open if every one of its VF-fibers

The expansion $\mathbb{U}^\bullet$ can help reduce the technical complexity of our discussion. However, as is the case with the definable Γ-sort, it is conceptually inessential since, for the purpose of this paper, all allusions to discs as (imaginary) elements may be eliminated in favor of objects already definable in $\mathbb{U}$.

Whether parameters in DC are used or not shall be indicated explicitly, if it is necessary, by means of introducing parameters of the form $\ulcorner \mathfrak{a} \urcorner$ or the phrase "in $\mathbb{U}^\bullet$," in which case the substructure $\mathbb{S}$ may contain names for discs that may or may not be definable from $\mathrm{VF}(\mathbb{S}) \cup \mathrm{RV}(\mathbb{S})$.

A disc centered at 0 may be identified with its valuative radius together with its open/closed designation (Γ is already treated as a definable sort). At a fixed finite radius the open and closed discs are distinct; the radius ∞ designates only the closed singleton disc.

For a disc $\mathfrak{a} \subseteq \mathrm{VF}$, the corresponding imaginary element in DC is denoted by $\ulcorner \mathfrak{a} \urcorner$ when notational distinction makes the discussion more streamlined; $\ulcorner \mathfrak{a} \urcorner$ may be heuristically thought of as the "name" of $\mathfrak{a}$. Conversely, a set $D \subseteq \mathrm{DC}$ is often identified with the set $\{\mathfrak{a} \mid \ulcorner \mathfrak{a} \urcorner \in D\}$, in which case $\bigcup D$ denotes a subset of VF.

Resolved referee R024 [Clarification] – A uniform language for the disc sort.
Original source line 791. For a literal language expansion, use two uniform maps $(a, \gamma) \mapsto \mathfrak{o}(a, \gamma)$ and $(a, \gamma) \mapsto \mathfrak{c}(a, \gamma)$, with domain $\mathrm{VF} \times \Gamma$ (and the specified degenerate cases). A separate named function for every element γ of the monster would name those radii and change the parameter structure. Also retain an open/closed tag when identifying discs centered at 0 with radii: the two discs with a fixed finite radius are distinct.

Author resolution. The original wording was informal shorthand for uniform maps with the radius as an argument. The maps and the open/closed designation are now explicit. Open discs have finite valuative radius; only closed discs admit the singleton radius-∞ case.

2.2. **Of points, discs, and atoms.** The substructure $\mathbb{S}$ is VF-*generated* if it is generated solely by elements in VF, similarly for being $(\mathrm{VF}, \mathrm{RV})$-generated, and so on. If $\mathbb{S}$ is VF-generated then the map rv is surjective in $\mathbb{S}$, and the standing assumption makes $\Gamma(\mathbb{S})$ nontrivial. Then the model-theoretic algebraic closure $\operatorname{acl}\mathbb{S}$ of $\mathbb{S}$ is a model of $\mathrm{ACVF}(0,0)$; see, for instance, [16, Corollary 4.18]. In that case, we also know what the definable closure $\operatorname{dcl}\mathbb{S}$ of $\mathbb{S}$ is:

Lemma 2.11. *Suppose that* $\mathbb{S}$ *is* VF*-generated and* $\Gamma(\mathbb{S})$ *is nontrivial. Then* $\operatorname{dcl}\mathbb{S} = \mathbb{S}$ *if and only if the underlying valued field* $(\mathrm{VF}(\mathbb{S}), \mathcal{O}(\mathbb{S}))$ *of* $\mathbb{S}$ *is henselian.*

Proof. We have $\mathrm{VF}(\operatorname{acl}\mathbb{S}) = \mathrm{VF}(\mathbb{S})^{\mathrm{ac}}$, where the latter is the field-theoretic algebraic closure of $\mathrm{VF}(\mathbb{S})$. Since the valued field automorphisms of $(\mathrm{VF}(\mathbb{S})^{\mathrm{ac}}, \mathcal{O}(\operatorname{acl}\mathbb{S}))$ over $(\mathrm{VF}(\mathbb{S}), \mathcal{O}(\mathbb{S}))$ are in one-to-one correspondence with the $\mathcal{L}_{\mathrm{RV}}$-automorphisms of $\operatorname{acl}\mathbb{S}$ over $\mathbb{S}$ and $\mathrm{VF}(\mathbb{S})$ is the fixed field with respect to these automorphisms if and only if $(\mathrm{VF}(\mathbb{S}), \mathcal{O}(\mathbb{S}))$ is henselian, we see that $\mathrm{VF}(\operatorname{dcl}\mathbb{S}) = \mathrm{VF}(\mathbb{S})$ if and only if $(\mathrm{VF}(\mathbb{S}), \mathcal{O}(\mathbb{S}))$ is henselian. By quantifier elimination and routine syntactical inspection, $\mathrm{RV}(\operatorname{dcl}\mathbb{S}) = \mathrm{RV}(\mathbb{S})$. The lemma follows. □

We shall impose at a later stage the assumption that $\mathbb{S}$ is VF-generated and (without loss of generality) definably closed, that is, $\operatorname{dcl}\mathbb{S} = \mathbb{S}$.

Lemma 2.12 ([16, Lemma 4.10])**.** *For any finite definable set* A *in* VF *there is a definable bijection* $f : A \longrightarrow U$ *for some set* U *in* RV*.*

The phenomenon exemplified by this simple lemma is dubbed "internalizing finite sets." Its failure in positive residue characteristics belies the deep obstacles one faces in developing a Hrushovski-Kazhdan integration theory therein; see the discussion around [8, Lemma 3.9].

Proof. Let $\alpha = \mathrm{rad}(\mathfrak{a})$. Then, for any $a \in \mathfrak{a}$, b is algebraic over (a, α) and hence, by Lemma 2.13, is algebraic over a. Since there is an $a \in \mathfrak{a}$ such that $a_i \notin \mathrm{acl}(a_{\tilde{i}}, b)$ for all $i \in [n]$, by Lemma 2.14, b must be algebraic. □

Lemma 2.19 ([16, Lemma 4.15])**.** *If $\mathbb{S}$ is $(\mathrm{VF}, \mathrm{RV}, \Gamma)$-generated then every algebraic (respectively, definable) closed disc contains an algebraic (respectively, definable) point.*

Lemma 2.20 ([16, Lemma 4.17])**.** *If $\mathbb{S}$ is (VF, Γ)-generated then every algebraic (respectively, definable) disc contains an algebraic (respectively, definable) point.*

A pillar of the structure of definable sets in $\mathbb{U}$ is *C-minimality*, meaning that every definable subset of VF is a finite boolean combination of valuative discs, allowing additional parameters to name the individual discs.

Resolved referee R027 [Clarification] – Parameters of the balls in C-minimality.
Original source line 859. The balls in a finite Boolean decomposition need not be individually $\mathbb{S}$-definable. For example, a definable pair of conjugate algebraic points need not split into two definable singleton balls. Say 'balls with additional parameters (whose finite canonical family is algebraic over the defining parameters)' and retain the Galois-invariant family where individual components cannot be chosen.

Author resolution. Unqualified definability remains relative to the fixed ground parameter set. The individual discs in a finite Boolean decomposition may require additional parameters to be named; the sentence now makes that distinction explicit.

Corollary 2.21. *Suppose that $\mathbb{S}$ is $(\mathrm{VF}, \mathrm{RV}, \Gamma)$-generated. Let $\mathfrak{a} \subseteq \mathrm{VF}$ be a definable disc and A a definable subset of* VF*. If $\mathfrak{a} \cap A$ is a nonempty proper subset of $\mathfrak{a}$ then $\mathfrak{a}$ contains a definable point.*

Proof. It is not hard to see that, by C-minimality, if $\mathfrak{a} \cap A$ is a nonempty proper subset of $\mathfrak{a}$ then $\mathfrak{a}$ contains a definable closed disc and hence the claim is immediate by Lemma 2.19. □

Definition 2.22. Let D be a set of parameters. We say that a (not necessarily definable) nonempty set A *generates a (complete) D-type* if, for every D-definable set B, either $A \subseteq B$ or $A \cap B = \emptyset$. In that case, A is *D-type-definable* if no set properly contains A and also generates a D-type. If A is D-definable and generates a D-type, or equivalently, if A is both D-definable and D-type-definable then we say that A is *D-atomic* or *atomic over D*.

We simply say "atomic" when $D = \emptyset$.

In the literature, a type could be a partial type and hence a type-definable set may have nontrivial intersection with a definable set. In this paper, since partial types do not play a role, we shall not carry the superfluous qualifier "complete" in our terminology.

Lemma 2.23 ([18, Lemma 3.3])**.** *Suppose that $\Gamma(\mathbb{S})$ is nontrivial. Let A be an atomic set. Then A is γ-atomic for all $\gamma \in \Gamma$.*

Resolved referee R028 [Error] – Atomic base change needs a nontrivial base value group.
Original source line 879. The unrestricted statement remains false over a trivially valued base, as the human referee observed. Over a trivially valued algebraically closed field, $A = \mathcal{M} \setminus \{0\}$ is atomic; a parameter $\gamma > 0$ cuts it into the nonempty proper subset $\{x : 0 < \mathrm{val}(x) < \gamma\}$. Impose a nontrivial base value group here and propagate that assumption through uses of this lemma. In the cited proof, the essential step is that a nonempty definable subset of Γ contains a definable point; this fails for $(0, \infty)$ over the trivial base.

Author resolution. The author adopts nontrivial $\Gamma(\mathbb{S})$ as a standing assumption throughout the paper, matching the blanket hypothesis in [18]. It persists under parameter extension and supplies the

Corollary 5.24. *In $\tilde{\mathbb{C}}$ with $\mathbb{S} = \mathbb{C}(\!(t)\!)$, if f is a morphism of varieties over $\mathbb{C}$ with $\mathrm{Vol}_b([A_b]) = 0$ for all $b \in B$ then $\mathrm{Vol}([A]) = 0$.*

6. Special covariant bijections

From here on, we also assume that the substructure $\mathbb{S}$ of parameters is VF-generated. So every definable disc contains a definable point (Lemma 2.20). Another simple fact we shall occasionally use is that if $\gamma \in \Gamma$ is definable then there is a $\gamma' \geq \gamma$ such that $\gamma'^{\sharp}$ contains a definable point.

6.1. Invariance and covariance.

Definition 6.1. For $\gamma \in \Gamma$, let $\pi_\gamma : \mathrm{VF} \longrightarrow \mathrm{VF}/\mathcal{M}_\gamma$ be the quotient map. If $\gamma = (\gamma_1, \ldots, \gamma_n) \in \Gamma^n$ then π_γ denotes the product of the maps π_{γ_i}.

Let $\alpha \in \Gamma^n$ and $\beta \in \Gamma^m$. We say that a function $f : A \longrightarrow B$ with $A \subseteq \mathrm{VF}^n$ and $B \subseteq \mathrm{VF}^m$ is *(α, β)-covariant* if it (π_α, π_β)-contracts to a function $f_\downarrow : \pi_\alpha(A) \longrightarrow \pi_\beta(B)$, that is, $\pi_\beta \circ f = f_\downarrow \circ \pi_\alpha$. For simplicity, we shall often suppress parameters in notation and refer to (α, β)-covariant functions as $(\alpha, -)$-covariant or $(-, \beta)$-covariant or just covariant functions.

A set in VF is *α-invariant* if its characteristic function is $(\alpha, 0)$-covariant.

More generally, for sets A, B with RV-coordinates, a function $f : A \longrightarrow B$ is *covariant* if every one of its VF-fibers is covariant (recall Terminology 2.6). Similarly, a set is *invariant* if every one of its VF-fibers is invariant.

Definition 6.2. A definable function $f : A \longrightarrow B$ is *proper covariant* if

- the sets A_{VF}, $f(A)_{\mathrm{VF}}$ are bounded and the sets A_{RV}, $f(A)_{\mathrm{RV}}$ are doubly bounded,
- for each VF-fiber f_t of f there is a t-definable tuple $(\alpha_t, \beta_t) \in \Gamma$ such that f_t is (α_t, β_t)-covariant, $\mathrm{dom}(f_t)$ is α_t-invariant, and $\mathrm{ran}(f_t)$ is β_t-invariant.

A definable set A is *proper invariant* if A_{VF} is bounded, A_{RV} is doubly bounded, and for each VF-fiber A_t of A there is a t-definable tuple $\alpha_t \in \Gamma$ such that A_t is α_t-invariant.

Note that covariance makes sense even when the domain or the range of the function f has no VF-coordinates, in which case the contraction $f_\downarrow$ would be the empty function. In particular, if B is a subset of RV^m or even $\mathcal{P}(\mathrm{RV}^m)$ then f being proper covariant simply means that every $f^{-1}(x)$ is proper invariant. Then A being proper invariant may also be construed as the condition that its projection into the RV-coordinates is proper covariant.

Referee R098 [Error] – Fiberwise invariance does not supply global bounds.
Original source line 2008. The asserted characterization drops the global boundedness conditions in Definition 6.2. For example, let $A = 1^\sharp \times \mathrm{RV}$ and let $f : A \longrightarrow \mathrm{RV}$ be the second projection. Every fiber $f^{-1}(t)$ is proper invariant (over t), but $A_{\mathrm{RV}} = \mathrm{RV}$ is not doubly bounded, so f is not proper covariant. Retain the boundedness requirements explicitly, or formulate this as a characterization under those requirements. Also, when the target has no VF-coordinates the relevant contraction is a map to the singleton empty tuple; it is not the empty function unless its domain is empty. The extension to powerset-valued functions should specify separately which boundedness condition is imposed on their RV-valued relation.

Example 6.3. A proper invariant set is not required to be doubly bounded. However, if $\boldsymbol{U} \in \mathrm{RV}^{\mathfrak{db}}[*]$ then $\mathbb{L}\boldsymbol{U}$ is indeed a doubly bounded proper invariant set.

Remark 6.4. In defining proper covariance above, if A is closed then there is no need to demand that $f(A)_{\mathrm{RV}}$ be doubly bounded, because it is implied by the other conditions, as follows. There is an $\mathfrak{o}$-partition $p : A \longrightarrow \Gamma$ such that, for each open polydisc $\mathfrak{b}$ in question, that is, for each open polydisc $\mathfrak{b} \subseteq A$ of the form $\mathfrak{o}(a, p(a))$, $f(\mathfrak{b})$ lies in the range of some VF-fiber of f. By Lemma 2.53,